Homework Solutions

Math 1A

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题 Q1

Suppose f is a function and L and a are real numbers.

解. See sub-problems for individual limit computations.

Answer:

$$\boxed{\text{See sub-problems}}$$

(a) $\lim_{x\to a} f(x) = L$ means: for every $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < |x-a| < \delta \Rightarrow |f(x)-L| < \varepsilon$. Informally: we can make $f(x)$ as close to L as we like by taking x sufficiently close to (but not equal to) a. $\boxed{\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x-a| < \delta \Rightarrow |f(x)-L| < \varepsilon}$

(b) This requires graphing, which is a visual exercise. $\boxed{\text{See graph}}$

(c) This requires graphing, which is a visual exercise. $\boxed{\text{See graph}}$

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题 Q2

Is ∞ a number? What does $\lim_{x\to a} f(x) = \infty$ mean? Does $\lim_{x\to a} f(x) = \infty$ mean that the limit exists?

解. ∞ is **not** a real number.

$\lim_{x\to a} f(x) = \infty$ means that $f(x)$ grows without bound as $x \to a$.

The limit does **not exist** in the usual sense; we say the limit "equals infinity" as a shorthand for divergence.

Answer:

$$\boxed{\infty \text{ is not a number. The limit does not exist.}}$$

题 Q3

What does the Squeeze Theorem say? Draw a picture illustrating this theorem.

解. **Squeeze Theorem:** If $g(x) \le f(x) \le h(x)$ for all x near a (except possibly at a), and $\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L$, then $\lim_{x\to a} f(x) = L$.

Intuitively: if f is "squeezed" between two functions that both converge to L, then f must also converge to L.

Answer:

$$\boxed{\text{If } g(x) \le f(x) \le h(x) \text{ and } \lim g = \lim h = L, \text{ then } \lim f = L.}$$

题 Q4

What do we mean by $\lim_{x\to a^-} f(x) = L$? What is meant by $\lim_{x\to a^+} f(x) = L$? Graph a function $y = f(x)$ and some point $x = a$ where $\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$. Find an algebraic expression for such a function.

解. $\lim_{x\to a^-} f(x) = L$: the **left-hand limit**. As x approaches a from values *less than* a, $f(x)$ approaches L.

$\lim_{x\to a^+} f(x) = L$: the **right-hand limit**. As x approaches a from values *greater than* a, $f(x)$ approaches L.

The two-sided limit $\lim_{x\to a} f(x) = L$ exists if and only if both one-sided limits exist and are equal.

Example: $f(x) = \begin{cases} x & x < 0 \\ x+1 & x \ge 0 \end{cases}$ at $a = 0$: $\lim_{x\to 0^-} f(x) = 0 \neq 1 = \lim_{x\to 0^+} f(x)$.

Answer:

$$\boxed{\lim_{x\to a^-} f(x) : \text{from below}; \quad \lim_{x\to a^+} f(x) : \text{from above}}$$

题 P1

(a) Is the limit of a sum always the sum of the limits? Give an example where $\lim_{x\to a}[f(x) + g(x)]$ exists even though neither $\lim_{x\to a} f(x)$ nor $\lim_{x\to a} g(x)$ exist.

解. No, the limit of a sum is *not* always the sum of the limits.

The **Sum Law** requires both individual limits to exist: $\lim[f(x)+g(x)] = \lim f(x) + \lim g(x)$ *if both limits exist.*

Counterexample: Let $f(x) = \frac{1}{x}$ and $g(x) = -\frac{1}{x}$. Neither $\lim_{x\to 0} f(x)$ nor $\lim_{x\to 0} g(x)$ exists, but $\lim_{x\to 0}[f(x) + g(x)] = \lim_{x\to 0} 0 = 0$.

Answer:

$$\boxed{\text{No. } f(x) = 1/x,\ g(x) = -1/x : \lim[f+g] = 0 \text{ but } \lim f, \lim g \text{ DNE}}$$

题 P2

(a) Find $\lim_{x\to 0^+}(\frac{1}{x} - \frac{1}{|x|})$, $\lim_{x\to 0^-}(\frac{1}{x} - \frac{1}{|x|})$, and $\lim_{x\to 0}(\frac{1}{x} - \frac{1}{|x|})$, if they exist.

解. $\lim_{x\to 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$
$\lim_{x\to 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$
$\lim_{x\to 0} -\frac{1}{|x|} + \frac{1}{x} = 0$
Answer:

$$\boxed{0}$$

(b) Computing: $\lim_{x\to 0} |x| = 0$ This confirms the statement. □$\boxed{0}$

(c) Left limit: $\lim_{x\to 0^-} \frac{|x|}{x} = -1$ Right limit: $\lim_{x\to 0^+} \frac{|x|}{x} = 1$ Since $-1 \neq 1$, the limit does not exist. $\boxed{\text{DNE}}$

题 P3

For what kinds of functions can one evaluate the limit $\lim_{x\to a} f(x)$ by just plugging in $x = a$? Give an example of a function $f(x)$ for which the limit $\lim_{x\to a} f(x)$ exists and can be evaluated, yet for which this substitution doesn't work.

解. **Continuous functions** allow direct substitution: $\lim_{x\to a} f(x) = f(a)$.

This includes polynomials, rational functions (at non-zero denominator points), trigonometric, exponential, and logarithmic functions at points in their domain.

Example where limit exists but substitution fails: $f(x) = \frac{x^2-1}{x-1}$ at $x = 1$. Direct substitution gives $\frac{0}{0}$, but $\lim_{x\to 1} \frac{x^2-1}{x-1} = \lim_{x\to 1}(x+1) = 2$.

Answer:

$$\boxed{\text{Continuous functions: } \lim_{x\to a} f(x) = f(a)}$$

题 AP1

Below are listed three limits. Match the best method to each limit and evaluate each limit.

解. See sub-problems for individual limit computations.

Answer:

$$\boxed{\text{See sub-problems}}$$

(a) $\lim_{x\to 0} \frac{x^{1000}-703x^{506}+\pi x^{17}+480}{-372x^{75}+39x^{14}+\sqrt{7}x^5-24} = -20$ $\boxed{-20}$

(b) $\lim_{x\to 4} \frac{x^2+2x-24}{x-4} = 10$ $\boxed{10}$

(c) $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0$ $\boxed{0}$

题 AP2

Using a calculator, try to guess $\lim_{n\to\infty} n\sin(\frac{\pi}{n})$. (Remember that n is in radians, not degrees!)

解. $\lim_{x\to\infty} n\sin\left(\frac{\pi}{n}\right) = \pi$

Answer:

$$\boxed{\pi}$$

题 AP3

Draw a circle of radius 1. Inscribe a regular n-gon inside it.

解. 半径为 1 的圆内接正 n 边形，其顶点把圆心角等分为 n 份。

因此每个中心角为 $\frac{2\pi}{n}$（弧度），即 $\frac{360^\circ}{n}$。

取相邻两个顶点与圆心构成一个等腰三角形，两腰长为半径 1，夹角为 $\frac{2\pi}{n}$。

该三角形面积为 $\frac{1}{2}\cdot 1\cdot 1\cdot \sin\frac{2\pi}{n} = \frac{1}{2}\sin\frac{2\pi}{n}$。

正 n 边形由 n 个这样的三角形组成，所以面积为 $A_n = n\cdot\frac{1}{2}\sin\frac{2\pi}{n} = \frac{n}{2}\sin\frac{2\pi}{n}$。

求极限：令 $x = \frac{2\pi}{n}\to 0$，则 $A_n = \pi\cdot\frac{\sin x}{x}\to\pi$。

直观上，当 n 越来越大时，正 n 边形越来越接近半径为 1 的圆，其面积应趋于圆的面积 π，与计算结果一致。

Answer:

$$\boxed{\frac{2\pi}{n}\ (\text{即}\frac{360^\circ}{n});\quad S_\triangle = \frac{1}{2}\sin\frac{2\pi}{n},\quad S_n = \frac{n}{2}\sin\frac{2\pi}{n};\quad \lim_{n\to\infty} S_n = \pi}$$

Auto-generated: 10/10 problems solved by SymPy.